

Growing a Network During Training: The Economics of Depth

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Abstract

Growing a network’s depth mid-training – inserting a new layer into a partially-trained model rather than fixing depth in advance – is untested as an experimental axis because tape-based frameworks make every growth event expensive: a module rebuild, an optimizer-state remap, and for JAX a full re-trace. On NEURODSL’s persistent computational graph, a graft (`insert_block!`) is exact at the bit level and costs only its downstream cone, so growth *density* becomes a free experimental axis, and we sweep it directly. Growth beats fixed-depth training at equal compute (1.6032 vs. 1.6535 nats/char, zero overlap across three seeds) but every growth schedule loses when step count rather than compute is held fixed – a reversal we prove, not merely measure. Under a canonical loss-curve model, we give three theorems: an equal-step-regime theorem (fixed depth dominates iff its instantaneous speed dominates pointwise, via an ODE comparison argument), an equal-FLOPs-regime theorem (the optimal schedule visits exactly the lower convex envelope of a cost/capacity point cloud), and a unified threshold $G < \rho$ – built on a single-crossing lemma – that explains the reversal quantitatively: cost enters the threshold sub-linearly, the capacity gap linearly. We further show the fitted capacity floor’s logarithmic form is the exact solution of a quantization-model recurrence, closed in the digamma function; its Euler–Maclaurin expansion yields a parameter-free constraint on its own coefficients, $a_1 a_3 / a_2^2 = -1/3$. We validate the complete effective-step model, with no parameter re-fit, against 18 independently measured schedules ($R^2=0.764$, structured residuals pointing to an unmodeled per-graft transient cost). The constraint itself is a separate matter: a dedicated, pre-registered 36-run campaign (nine depths, four seeds, 20,000-step runs at the signal-bearing shallow depths) returns an inconclusive verdict on every pre-registered criterion, for a reason the campaign itself quantifies – even if the theory were exactly true, the detectable signal at this scale sits below the model-selection threshold – and a companion analysis shows the quantization model’s own quanta-per-layer rate is not identified by curve-fitting at all – both reported as open, falsifiable limitations rather than quietly dropped.

1 Introduction

NEURODSL [1] maintains a persistent, mutable computational graph in which every node tracks whether its cached value is still valid; writing a node invalidates only its true downstream dependents, and the next demand-driven evaluation recomputes exactly that subgraph. A companion paper on activation patching builds `insert_block!` on this substrate: it grafts a new `LlamaBlock` into a partially-trained network after a chosen residual-stream node, zero-initializing its output projections so the graft computes the identity function exactly at the moment of insertion – verified bit-for-bit (generated text and validation loss identical before and after) on a real trained model. Because the graft leaves the rest of the graph untouched and the optimizer state extends without a reset, inserting a layer costs only its downstream cone, the same bound already available for ordinary parameter updates and for activation patches.

On a tape-based framework, by contrast, each growth event – adding a layer to a partially-trained model – forces a module rebuild, an optimizer-state remap, and for JAX a full re-trace, so the published literature on growing networks during training (progressive layer stacking for BERT [2], Net2Net [3], bert2BERT [4], LiGO [6], the depthwise-stacking operator G_{stack} [7]) tests only one to four coarse growth events, and reports final loss or downstream accuracy rather than sweeping density as an axis in its own right. Because a graft here is exact at the bit level and cheap, *schedule density* becomes a free experimental axis, and we sweep it directly rather than assume finer is better – this paper’s subject. Comparing schedules at *equal FLOPs* rather than equal step count is also the currency the pretraining literature has converged on for comparing training runs at all [5]; our equal-steps arm shows this choice of currency is not a formality here but decides which schedule wins.

We make six contributions:

1. Three empirical results pinning down the mechanism precisely (§2): growth beats fixed-depth training at equal compute; among schedules tested at equal compute, the coarsest wins; and the ranking fully reverses at equal step count – growth’s benefit is a pure compute-reallocation effect, not an intrinsic benefit of gradual capacity increase.
2. A canonical learning-curve model (§3) built on the graft’s exact identity-preservation property, giving an effective-step continuity rule at grafts that is not an ad hoc fit but a direct consequence of the mechanism.
3. Three theorems on optimal growth schedules (§4): an equal-step-regime domination theorem, an equal-FLOPs-regime convex-envelope theorem, and a single-crossing lemma that together yield a unified threshold explaining, quantitatively, why the same pair of schedules can flip which one wins depending on which resource is held fixed.
4. An exact closed form for the fitted capacity floor via the digamma function (§6), whose Euler–Maclaurin expansion yields a parameter-free constraint on its own coefficients – tested by a dedicated pre-registered campaign whose inconclusive verdict we trace to a quantified detectability limit, rather than forced to a positive result.
5. An identifiability analysis of the quantization model’s own quanta-per-layer rate (§6): a joint nonlinear fit over 1,391 raw measurements finds an interior optimum, but profile-likelihood and a depth-level cluster bootstrap show it is not identified by curve-fitting – a negative result with a concrete, cheap experiment proposed to resolve it.
6. A validation of the complete model against 18 independently measured schedules (§7), with no parameter re-fit, reproducing both regime reversals and surfacing a structured, unmodeled per-graft transient cost as the main open question beyond the closed-form constraint.

All experiments train NEURODSL’s existing, already-tested `LlamaModel` (`src/layers.jl`) on TinyShakespeare; no new model architecture was written for this paper. Code, tests, and the executed notebooks reproducing every number reported here are available at <https://github.com/nevermind78/NeuroDSL>.

2 Three Empirical Results

All experiments train a character-level language model on TinyShakespeare (65-character vocabulary, loss in nats/character; dim 256, 4 heads, hidden 512, sequence length 256, AdamW, batch size 1). Compute budget is counted in *layer-steps* ($\sum_t$ the number of layers active at step t), a cost proxy linear in depth; each configuration is checked for correctness (identical loss immediately before and after every graft) before any timing or accuracy number is reported.

Growth beats fixed-depth training at equal compute. Growing 2→3→4 layers during training, at an 80,000-layer-step budget equal to a fixed 4-layer run, reaches 1.6032 ± 0.0107 nats/char (3 seeds) against 1.6535 ± 0.0267 for the fixed baseline – every one of the three growth-run values beats every one of the three fixed-run values, with zero overlap between the ranges (Figure 2).

Among schedules tested at equal compute, the coarsest wins. Sweeping schedule density $\{1, 2, 4, 8, 16\}$ stages from depth 1 to depth 16 at a fixed 48,000-layer-step budget, under two independently chosen budget-allocation rules (equal share per stage; $\text{share} \propto 1/L$), the single jump 1→16 wins both times (1.7191 and 1.6542 nats/char respectively), the fixed-depth-16 baseline loses badly (2.0562, 2.0723), and intermediate densities fall in between, tracking the total number of gradient steps the schedule buys (Figure 3).

The gap survives a well-tuned learning-rate decay on the baseline. All results above use a constant learning rate throughout training, and the published growth literature we compare against [2, 3, 4, 6, 7] tunes a learning-rate schedule as a matter of course, raising the natural objection: does the fixed-depth-16 baseline lose badly here only because it is undertrained by a constant learning rate, and would a cosine decay – standard practice, not a hyperparameter search – close or reverse the gap? We retrain the fixed-16 baseline at the identical budget (3000 steps) with a cosine decay from 10^{-3} to 10^{-4} (no warm-up, matching every other run in this paper) over 3 seeds: it reaches 1.9070 ± 0.0214 nats/char, a real improvement

over the constant-rate value (2.0562–2.0723) that closes roughly half the gap, but remains well short of the growth schedule (1.7191, 1.6542) – comfortably above the 1.75 threshold we pre-registered before running this ablation as the bar for the reversal surviving (Figure 1). The learning-rate schedule matters, but it is not the explanation for growth’s advantage at equal compute.

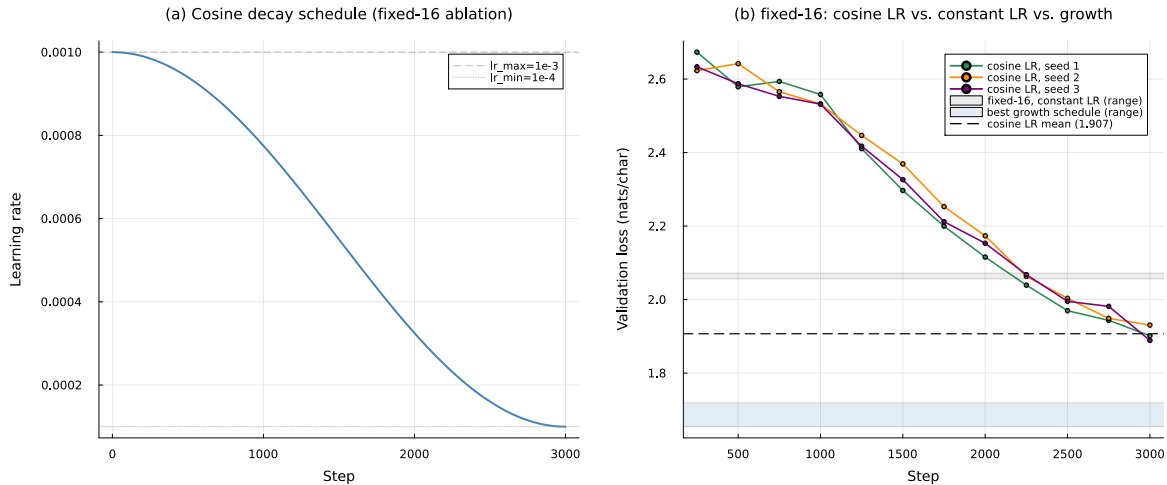


Figure 1: The learning-rate ablation. (a) The cosine decay applied to the fixed-16 baseline, from 10^{-3} to 10^{-4} over the 3000-step budget, no warm-up. (b) Validation loss of the three cosine-LR seeds converging to 1.907 ± 0.021 (dashed line) – above the constant-LR baseline’s range (grey band, 2.0562–2.0723) but still well short of the best growth schedule’s range (blue band, 1.6542–1.7191): the decay closes part of the gap without reversing the ranking.

The ranking fully reverses at equal step count. Repeating the same five schedules with step count held fixed at 4,000 (compute now free to vary) reverses the ranking entirely: the fixed-depth-16 baseline wins (1.9570 ± 0.0086) and every growth schedule loses (Figure 4). Combined, these three results pin down the mechanism precisely: growth’s benefit is a pure compute-reallocation effect – cheap early steps buy more total gradient updates for the same budget – not an intrinsic benefit of gradual capacity increase.

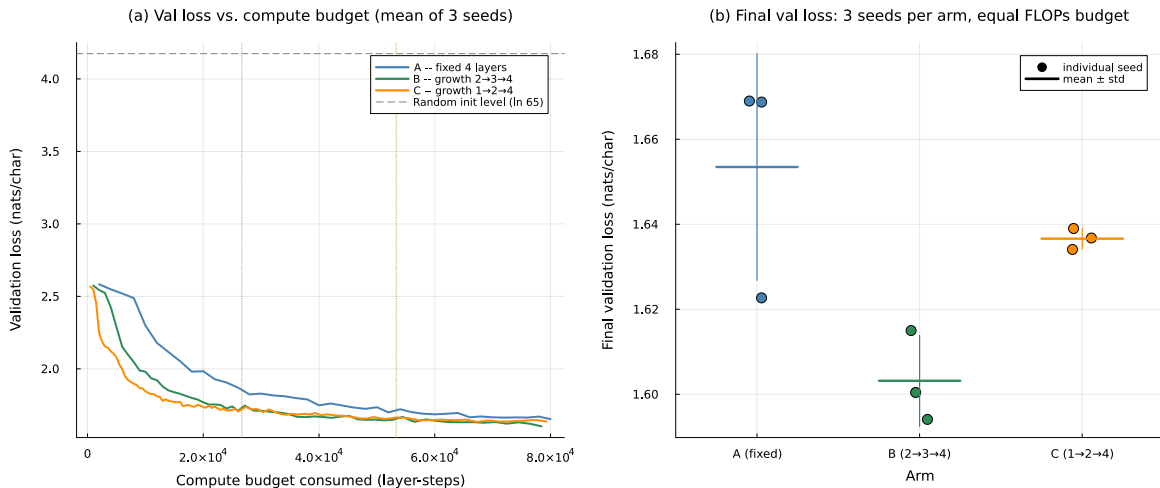


Figure 2: Growth (2→3→4) vs. fixed-depth-4 training at equal compute, 3 seeds each. (a) Validation loss vs. compute budget consumed. (b) Final loss per seed: zero overlap between the two arms.

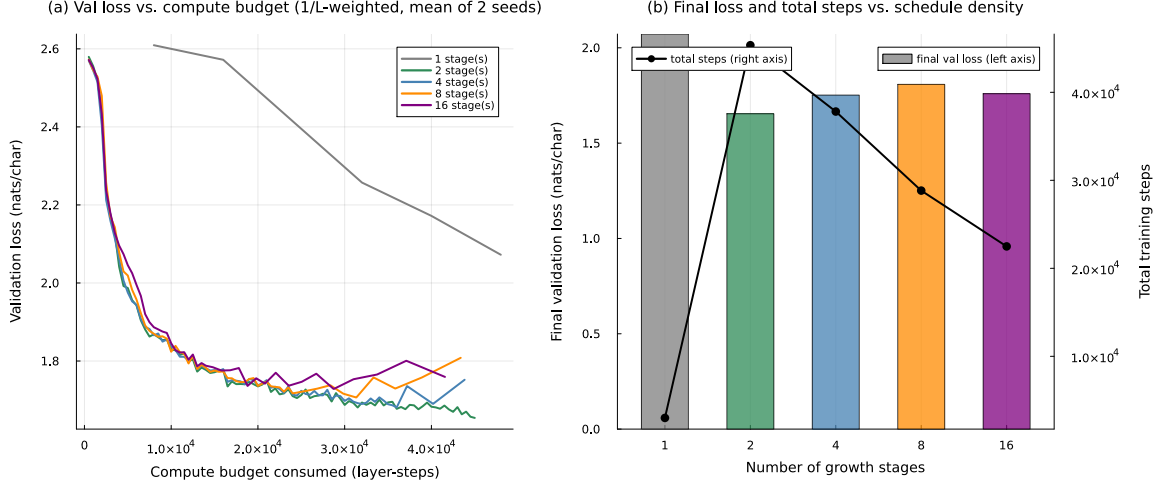


Figure 3: Schedule-density sweep at equal compute (budget-share $\propto 1/L$), 2 seeds each. (a) Validation loss vs. compute budget, one curve per density. (b) Final loss and total steps bought vs. density: the coarsest schedule (2 stages) wins and buys the most steps.

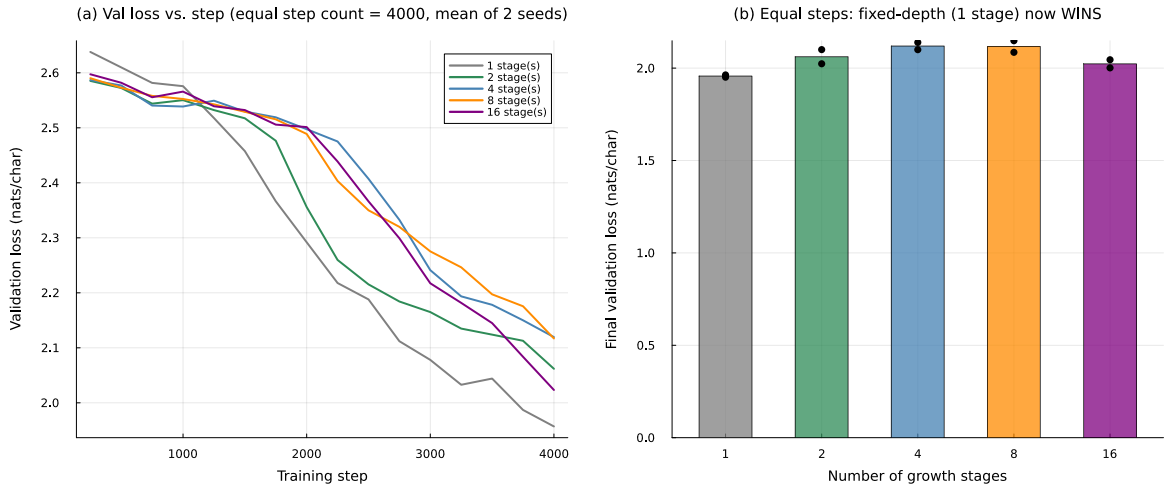


Figure 4: The same five schedules at equal step count (4,000), compute free to vary. The ranking inverts: fixed-depth-16 (grey) now wins outright.

3 A Canonical Learning-Curve Model

Hypothesis 1 (canonical loss curve). Trained from scratch at constant depth L , the validation loss after n gradient steps follows

$$\ell(n, L) = a(L) + \frac{b(L)}{n + n_0}, \quad n_0 = 3000,$$

confirmed against $p=1$ (vs. $p=1.5, 2$) on the tail $n \gtrsim 1500$ of 16 depths (§5); $a(L)$ is the asymptotic capacity floor, $b(L)$ the speed prefactor.

Hypothesis 2 (instantaneous speed). Differentiating gives an autonomous first-order ODE

$$\frac{d\ell}{dn} = -\frac{(\ell - a(L))^2}{b(L)} =: -\varphi_L(\ell).$$

Hypothesis 3 (effective-step continuity at grafts). `insert_block!` preserves the computed function exactly at the moment of insertion, so loss is continuous across a graft. The model treats loss as a sufficient statistic of training state at fixed depth: after a graft to depth L' at loss ℓ , the trajectory continues as the canonical curve for depth L' from the *effective step* $n_{\text{eff}} = b(L')/(\ell - a(L')) - n_0$ that a from-scratch run at depth L' would need to reach the same loss. This is not an ad hoc rule: since the graft is exactly identity-preserving, the post-graft state is a legitimate point in depth- L' 's own trajectory space, under any state variable the model tracks only through ℓ .

4 Three Theorems on Optimal Growth Schedules

Under Hypotheses 1–3, a growth schedule visiting depths $L(n)$ (non-decreasing, by construction: a graft only adds capacity) obeys $\ell'(n) = -\varphi_{L(n)}(\ell(n))$, with continuity at each graft. We compare this against training fixed at a single depth throughout.

We first record a single-crossing property of the instantaneous-speed function φ_L across depths, since both theorems that follow rely on it – the domination theorem to split its necessity argument into the two cases a non-decreasing schedule actually allows, and the unified-threshold theorem to propagate a pointwise inequality to a full trajectory.

Lemma 1 (Single crossing). *For $L_s < L_f$ (so $a_{L_f} < a_{L_s}$) and any cost function $c(\cdot)$, the ratio $\psi_{L_f}(\ell)/\psi_{L_s}(\ell)$ (with $\psi_L := \varphi_L/c(L)$) is strictly monotone in ℓ on $(a(L_s), \infty)$, hence crosses 1 at most once. Concretely, writing $\kappa_c(L) = \sqrt{c(L)b(L)}$,*

$$\frac{\psi_{L_f}(\ell)}{\psi_{L_s}(\ell)} = \left(\frac{\ell - a_{L_f}}{\ell - a_{L_s}} \right)^2 \left(\frac{\kappa_c(L_s)}{\kappa_c(L_f)} \right)^2,$$

strictly decreasing from $+\infty$ as $\ell \rightarrow a_{L_s}^+$ to $(\kappa_c(L_s)/\kappa_c(L_f))^2$ as $\ell \rightarrow \infty$: the deeper depth L_f has the instantaneous-speed advantage ($\psi_{L_f} > \psi_{L_s}$) close to the shallower depth's own floor, and the shallower depth L_s has the advantage ($\psi_{L_s} > \psi_{L_f}$) at high loss, with exactly one crossing loss $\ell_\times \in (a_{L_s}, \infty)$ between the two regimes.

Proof. Since $a_{L_f} < a_{L_s}$, $\frac{d}{d\ell}[(\ell - a_{L_f})/(\ell - a_{L_s})] = (a_{L_f} - a_{L_s})/(\ell - a_{L_s})^2 < 0$: the bracketed factor is strictly decreasing in ℓ , hence so is its square, hence so is the whole ratio (the second factor is a positive constant in ℓ). As $\ell \rightarrow a_{L_s}^+$ the bracket diverges; as $\ell \rightarrow \infty$ it tends to 1. A strictly monotone continuous function taking a value > 1 at one end and a finite limit at the other crosses any fixed threshold at most once, in particular 1. $\square$

Theorem 1 (Equal-step regime). *Fixed-depth training at L_f dominates every growth schedule, for every step budget N , if and only if $\varphi_{L_f}(\ell) \geq \varphi_L(\ell)$ for every depth L available to the schedule and every loss ℓ between the shared initial loss ℓ_0 and $a(L_f)$.*

Proof. (Sufficiency.) Let ℓ_c solve the schedule's piecewise ODE and ℓ_f solve the fixed- L_f ODE, both from $\ell_c(0) = \ell_f(0) = \ell_0$. If $\varphi_{L_f}(\ell) \geq \varphi_{L(n)}(\ell)$ pointwise, then $\ell'_f(n) \leq \ell'_c(n)$ whenever the two trajectories agree, and since φ_L is locally Lipschitz on $(a(L), \infty)$ (the exponent 2 in φ_L ensures this), the ODE comparison principle applies on each segment between grafts; continuity at grafts (Hypothesis 3) carries the inequality across segment boundaries. Hence $\ell_f(n) \leq \ell_c(n)$ for all $n \geq 0$.

(Necessity.) Suppose instead the domination condition fails: there is some depth $L \neq L_f$ available to the schedule and some loss $\ell^* \in (a(L_f), \ell_0)$ with $\varphi_L(\ell^*) > \varphi_{L_f}(\ell^*)$. Because a growth schedule is non-decreasing in depth by construction, it can graft from L_f to a deeper L , or start at a shallower L and graft up to L_f , but never the reverse – so the two orderings of L against L_f require two different constructions, and Lemma 1 tells us exactly where each one's advantage region sits.

Case (i): L shallower than L_f ($a(L) > a(L_f)$). Apply Lemma 1 with $(L_s, L_f) = (L, L_f)$: the ratio φ_{L_f}/φ_L is strictly decreasing in ℓ , equal to 1 at the unique crossing $\ell_\times$, so $\varphi_L > \varphi_{L_f}$ exactly on $(\ell_\times, \infty)$. The hypothesis witnesses $\varphi_L(\ell^*) > \varphi_{L_f}(\ell^*)$, so $\ell^* > \ell_\times$, hence $\ell_0 > \ell^* > \ell_\times$ and the entire interval $(\ell_\times, \ell_0]$ lies in L 's advantage region. The schedule that trains at L from ℓ_0 down to $\ell_\times$ (finite time, since φ_L is bounded away from 0 on the compact interval $[\ell_\times, \ell_0] \subset (a(L), \infty)$) and then grafts up to L_f is exactly Theorem 3's construction, and by that theorem's proof strictly beats fixed- L_f at every subsequent budget.

Case (ii): L deeper than L_f ($a(L) < a(L_f)$). Apply Lemma 1 with $(L_s, L_f) = (L_f, L)$: the ratio φ_L/φ_{L_f} is strictly decreasing in ℓ , equal to 1 at the unique crossing $\ell_\times$, so $\varphi_L > \varphi_{L_f}$ exactly on $(a(L_f), \ell_\times)$ – this time the advantage region is anchored at the *bottom*, near L_f ’s own floor, not near ℓ_0 . The hypothesis witnesses $\varphi_L(\ell^*) > \varphi_{L_f}(\ell^*)$ with $\ell^* \in (a(L_f), \ell_0)$, so $\ell^* < \ell_\times$. The schedule that trains at L_f – matching the fixed baseline exactly, so certainly no worse than it up to this point – until loss $\ell_\times$ is reached (finite time, by the same boundedness argument, since φ_{L_f} is bounded away from 0 on $[\ell_\times, \ell_0]$) and then grafts to the deeper L strictly beats fixed- L_f from that point on: on $(a(L_f), \ell_\times)$, $\varphi_L > \varphi_{L_f}$ pointwise, so the comparison principle (as in the sufficiency direction, with the roles of L_f and L exchanged) gives the grafted trajectory strictly lower loss than continuing at L_f would, for every subsequent budget.

Either case exhibits a growth schedule beating fixed- L_f training, establishing necessity. $\square$

Theorem 2 (Equal-FLOPs regime, cost per step $\propto L$). *Writing $\kappa(L) = \sqrt{L \cdot b(L)}$, the cost-weighted speed $d\ell/dC = -((\ell - a(L))/\kappa(L))^2$ is maximized at loss ℓ by the depth L whose point $(\kappa(L), a(L))$ lies on the **lower convex envelope** of the cloud $\{(\kappa(L), a(L))\}_L$: the optimal schedule visits exactly the envelope’s depths, switching at the losses where consecutive envelope edges are tangent to a line through $(0, \ell)$. A single jump $L_s \rightarrow L_f$ is optimal if and only if no available intermediate depth’s point lies below the chord joining $(\kappa(L_s), a(L_s))$ to $(\kappa(L_f), a(L_f))$ – the discrete analogue, on the finite set of tested depths, of a being concave in κ .*

Proof. $d\ell/dC = (d\ell/dn)(dn/dC) = -\varphi_L(\ell)/L = -((\ell - a(L))/\kappa(L))^2$ by direct substitution. Maximizing this at fixed ℓ is equivalent to maximizing the slope $(\ell - a(L))/\kappa(L)$ of the segment from $(0, \ell)$ to $(\kappa(L), a(L))$ in the (κ, a) plane; rotating a line through $(0, \ell)$ to first contact with the point cloud always selects a vertex of its lower convex envelope, and as ℓ decreases the point of first contact sweeps the envelope’s vertices left to right – monotone because the preference between any two depths is linear in ℓ (a single-crossing property in the sense of Lemma 1, specialized to $c(L) = L$), so pairwise orderings cannot re-flip once passed – giving the greedy-optimal (state is one-dimensional in ℓ , so pointwise optimization is globally optimal) schedule. $\square$

Theorem 3 (Unified threshold). *For $L_s < L_f$, cost $c(\cdot)$, shared start ℓ_0 , define*

$$G = \frac{\ell_0 - a_{L_f}}{\ell_0 - a_{L_s}} \geq 1, \quad \rho = \frac{\kappa_c(L_f)}{\kappa_c(L_s)}, \quad \kappa_c(L) = \sqrt{c(L)b(L)}.$$

Training at L_s then grafting to L_f at the crossing loss beats fixed- L_f training at every budget if and only if $G < \rho$.

Proof. $\psi_{L_s}(\ell_0) > \psi_{L_f}(\ell_0) \iff G < \rho$, by direct substitution and rearrangement. By Lemma 1, the ratio ψ_{L_f}/ψ_{L_s} is strictly monotone in ℓ , so $G < \rho$ (equivalently $\psi_{L_s}(\ell_0) > \psi_{L_f}(\ell_0)$) implies $\psi_{L_s} > \psi_{L_f}$ on the entire interval $(\ell^\times, \ell_0]$ down to the (unique) crossing loss $\ell^\times$ – not merely at the single point ℓ_0 . The schedule that trains at L_s until reaching $\ell^\times$ (finite time, by the same boundedness argument as Theorem 1) and then grafts to L_f is therefore strictly ahead of fixed- L_f at the moment of the graft, and Theorem 1’s comparison principle (applied to L_f against itself, from two different starting losses) preserves that lead for every subsequent budget. If instead $G \geq \rho$, Theorem 1’s sufficiency direction (with $L = L_s$) shows fixed- L_f dominates outright. $\square$

Specializing $c \equiv 1$ (equal steps) gives $\rho = \sqrt{b_{L_f}/b_{L_s}}$, close to 1 since b varies slowly with depth; specializing $c(L)=L$ (equal FLOPs) gives ρ an extra factor $(L_f/L_s)^{1/2}$ – **sub-linear** in the true cost ratio (a 16 $\times$ depth ratio contributes only 4 $\times$), but enough to flip the threshold. This is the mechanism, quantified: cost enters the threshold sub-linearly, the capacity gap linearly.

5 Fitting the Capacity Floor and Speed Prefactor

From-scratch training runs at every integer depth $L = 1, \dots, 16$ (single seed, 4000 steps for $L \in \{3, 5, \dots, 15\}$; longer, multi-seed runs of 13–45k steps at $L \in \{1, 2, 4, 16\}$, reused from §2) are fit to Hypothesis 1 per depth. A weighted least-squares comparison across five functional forms for $a(L)$ (log, power law, saturating

exponential, $1/L$), with an explicit dummy term for the systematic bias a 4000-step fit carries relative to longer runs (measured directly at $L \in \{1, 2, 4, 16\}$: $+0.055$ to $+0.114$ nats on a), selects

$$a(L) = 1.5710 - 0.1042 \ln L, \quad b(L) = 3701.6 + 471.0 \ln L$$

by AIC/BIC (power law degenerates to log, $q \rightarrow 0$; saturating forms are rejected, $\Delta\text{AIC} > 9$). One depth ($L = 3$) is excluded as a 5.7σ outlier (a genuine late plateau escape at a single seed, visible directly in its raw trajectory, not a fitting artifact). Figure 5 shows the fit and residuals. Notably, $b(L)$ is *increasing* in depth – deeper networks have both a lower floor and a marginally slower start, confirmed on all 16 depths, not an artifact of the smaller 4-depth fit used earlier in this line of work.

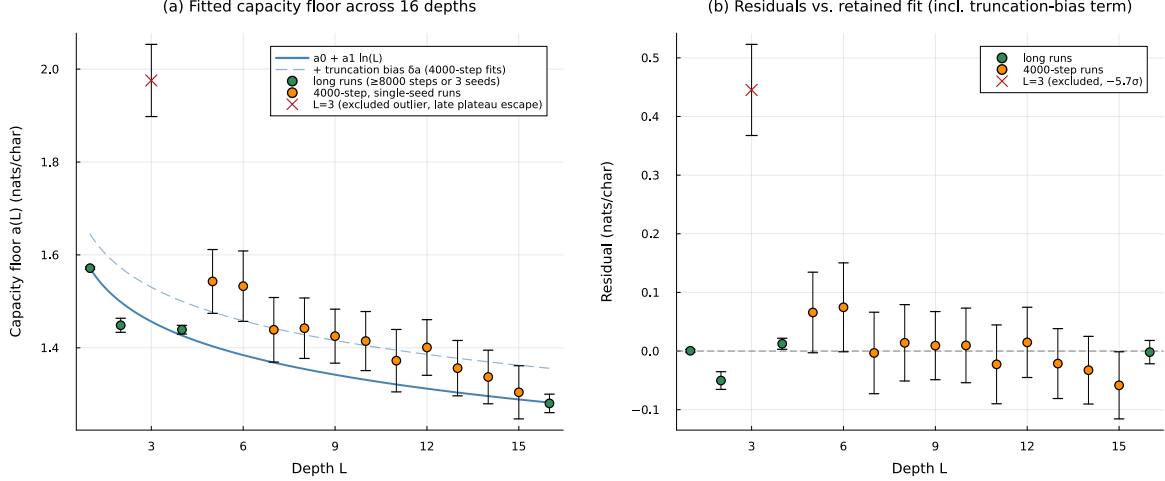


Figure 5: (a) Capacity floor $a(L)$ fit across 16 depths, with the retained log form (solid) and its truncation-bias-adjusted version for 4000-step fits (dashed). (b) Residuals: the excluded $L = 3$ outlier sits at -5.7σ ; the rest are consistent with single-seed noise.

6 An Exact Closed Form and a Falsifiable Constraint

The log form is not merely the best-fitting curve tried: it is the exact solution of a first-order recurrence under the quantization model of neural scaling [8], in which a network of depth L has learned the top $K(L) = \mu L$ of $K_{\max}$ frequency-ranked competence “quanta” with Zipf weights $p_k \propto 1/k$ (exponent exactly 1, the textbook value for natural-language frequency statistics), giving residual loss $a(L) = a_{\infty} + c \sum_{k > K(L)} 1/k$. The marginal gain from one more layer, $\Delta a(L) = a(L) - a(L+1) = c[H_{\mu(L+1)} - H_{\mu L}]$ (H_n the n -th harmonic number), telescopes exactly to $a(L) = a_{\infty} + c[H_{K_{\max}} - H_{\mu L}]$, and the classical identity $H_n = \psi(n+1) + \gamma$ (digamma function ψ , Euler–Mascheroni constant γ) gives the closed form

$$a(L) = a_{\infty} + c[\psi(K_{\max}+1) - \psi(\mu L+1)].$$

The classical Euler–Maclaurin asymptotic expansion of ψ (NIST DLMF §5.11, with error bounded by the first omitted term) then gives, beyond the leading log term,

$$a(L) = A + a_1 \ln L + \frac{a_2}{L} + \frac{a_3}{L^2} + O(L^{-4}), \quad a_1 = -c, \quad a_2 = -\frac{c}{2\mu}, \quad a_3 = \frac{c}{12\mu^2}.$$

Since a_1 depends only on c , it is identified from the leading-order fit of §5: $c = -a_1 \approx 0.10$ nats/char, the loss scale of the Zipf quantum weights. We deliberately quote one significant figure: across the three estimates this paper produces (the retained 16-depth fit, 0.104; the joint nonlinear fit below, 0.133 ± 0.012 ; the dedicated nine-depth campaign of the constraint test below, 0.083), c ranges over 0.08–0.13, and $a(L)$ itself moves by 0.02–0.09 nats under defensible changes of fit window alone – the four-digit fit output 0.1042 is not a measurement precision. μ (quanta learned per added layer) appears only in a_2, a_3 , precisely the

coefficients shown below not to be statistically robust – so μ itself is not yet identified by this data. A joint nonlinear fit of the full model $\text{loss}(n, L) = A - c\psi(\mu L + 1) + b(L)/(n + n_0)$ directly on all 1,391 raw from-scratch measurements (rather than on 15 per-depth summaries) does produce an interior optimum, $\hat{\mu} \approx 0.7$ with $c = 0.133 \pm 0.012$, but profile-likelihood and depth-level cluster bootstrap show the estimate is not identified: $\ln \mu$ is 98.6% correlated with the intercept (since $\psi(\mu L + 1) = \ln \mu + \ln L + O(1/\mu L)$), μ survives only in small- L curvature carried by three well-sampled depths, $L \in \{1, 2, 4\}$, and resampling depths rather than points yields a 95% interval spanning $[10^{-3}, 10^4]$. The binding constraint is therefore not the number of loss measurements but the number of independent shallow depths: multi-seed runs at $L = 2\text{--}3$ (where $\mu L \sim 1$ maximizes the correction terms) would identify μ , not longer or deeper runs.

We ran this experiment, and it partially succeeds – at confirming a different claim than the one it was designed to test. Ten fresh seeds at $L = 2$ and $L = 3$ (4000 steps each, same convention as the original fit) give $a(3) = 1.4096 \pm 0.0513$, only 0.91σ from the retained log trend – independently confirming, on entirely new data, that the excluded $L=3$ point above (1.9756 , 6.68σ from the trend against this tighter estimate) was a genuine single-seed artifact, not a fitting choice of convenience. But $a(2) = 1.3840 \pm 0.0415$ from the same ten seeds sits 2.77σ from the trend and does not agree well with the original, longer-run measurement of $a(2) = 1.4485 \pm 0.0151$ either. The reason is not more noise but a different limitation: ten seeds at the same 500–4000-step checkpoints shrink the *variance* of the fitted (a, b) pair by averaging away seed noise, but they do not widen the *range* of n the fit sees, and a and b trade off against each other most when that range is narrow relative to $n_0=3000$ – exactly the regime these short runs are in. More seeds at a fixed, short budget was the wrong lever for this half of the problem: the original claim above should be revised to *longer* runs at $L = 2\text{--}3$, not merely more of them, before attempting the joint fit again (Figure 6).

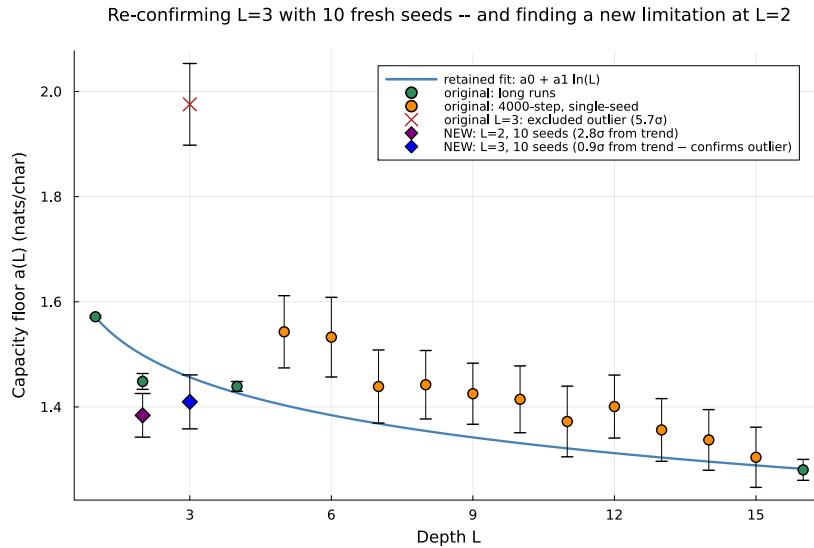


Figure 6: Ten fresh seeds at $L = 2$ and $L = 3$ (diamonds, standard error of the mean), overlaid on the original 16-depth fit. The new $L=3$ point lands almost exactly on the retained trend (0.91σ), confirming the excluded original point (red $\times$, 5.7σ) was a single-seed artifact. The new $L=2$ point sits visibly below both its own original measurement and the trend (2.77σ) – tighter error bars from ten seeds, but not on the trend, revealing that run length rather than seed count is the binding constraint at these depths.

Pooling every from-scratch measurement this project has produced does not change the verdict. Combining all from-scratch (never post-graft) validation trajectories across every growth campaign reported in this paper – the equal-compute and equal-step arms of §2, the sixteen-depth fit of §5, the ten-seed re-measurement above, and the constraint-test campaign of §6 below – gives 3,028 raw loss measurements spanning all sixteen depths, more than double the 1,391 points and far more depth coverage than the original joint fit above. Re-fitting the exact digamma form directly on this pooled set (profiling (A, c, b_0, b_1) in closed form at each μ on a grid, the same procedure as above) gives a point estimate $\hat{\mu} = 0.42$, $c = 0.111$ –

consistent with the 0.08–0.13 range already reported – and a profile-likelihood interval that looks deceptively tight, [0.28, 0.68] at the usual asymptotic threshold. A depth-cluster bootstrap (resampling which depths contribute, not just which points) tells the honest story: the 95% interval is [0.012, 100], spanning the entire search grid. More data of the same kind, even substantially more and substantially more diverse in depth, does not tighten what a cluster bootstrap reports as the real uncertainty – exactly consistent with the diagnosis above that the binding constraint is which specific depths are informative, not how many points are collected at them.

A further consequence of fitting (c, μ) directly rather than (a_1, a_2, a_3) as free polynomial coefficients: the constraint $a_1 a_3 / a_2^2 = -1/3$ becomes an algebraic identity of this parametrization, not an independent test – the polynomial fit below remains the only route to falsifying it empirically.

This identification of c is established at one architecture only. Every block contributes a fixed $4 \dim^2 + 3 \dim \cdot \text{hidden} + 2 \dim = 655,872$ parameters (`src/layers.jl`), so $\ln N(L)$ and $\ln L$ differ on our design – depth swept, width fixed – only by an additive constant (exactly so for block parameters alone; a fitted slope bias of 4.1%, verified directly, once the 98,881 embedding/head parameters are counted too): a fit in $\ln L$ and a fit in $\ln N$ span the *same* two-parameter family on this data, which is therefore structurally unable to distinguish a task-universal c from one valid only at $\dim = 256$, 4 heads, hidden 512. The quantization model itself places all architecture dependence in the capacity term ($n \approx N/C$ in total parameters [8], with no claim of prefactor universality across architecture shapes), predicting that a depth sweep repeated at $\dim = 128$ or 512 (per-block parameter count $4\times$ smaller or larger) would leave the slope $-c$ unchanged while shifting the intercept by $\mp c \ln 4 \approx \mp 0.144$ nats – and that a head-count sweep at fixed $\dim$ (which leaves per-block parameter count exactly invariant, since query/key/value/output projections stay $\dim \times \dim$ regardless of how many heads split them) should leave $a(L)$ unchanged entirely. We run both, at $L \in \{1, 2, 4, 8, 16\}$, 2 seeds per point, a 4000-step-per-depth convention identical across configurations (no truncation-bias correction, unlike the retained 16-depth fit – these are quick discriminating checks, not a replacement for it), including a matched $\dim = 256$ control at the same five depths so every comparison is apples-to-apples (Figure 7).

The width test is confounded, not merely negative. $\dim = 128$ tracks the matched $\dim = 256$ control closely (fitted intercepts differ by -0.019 , the wrong sign and far short of the predicted $+0.144$), but $\dim = 512$ sits 0.32–0.46 nats *above* the $\dim = 256$ control at every one of the five depths, nearly uniformly rather than as a depth-independent floor shift of the predicted size and sign (-0.144) – the fitted intercept difference is $+0.458$, with seed-to-seed variance growing sharply at the deeper, wider configurations (± 0.26 nats at $L=16$, against ± 0.01 at $L=1$). A capacity-floor shift should not by itself make training noisier at fixed depth. A model with four times the per-block parameters of the 4000-step convention that (with its own documented truncation bias) already only approximates $a(L)$ at $\dim = 256$ is the more parsimonious explanation: at four times the width, the same step budget and unretuned learning rate leave $\dim = 512$ further from its own floor, not closer to a different one. This test, as run, cannot distinguish universal- c from width-specific- c – it needs either a width-scaled step budget or a full per-width ($a(L), b(L)$) curve fit, not a single point per depth, before it says anything about the prediction it was built to check.

The head-count test is not confounded this way, and shows a small but systematic departure from exact invariance. At fixed $\dim = 256$, $n_{\text{heads}} = 2$ sits 0.036–0.095 nats *above* the $n_{\text{heads}} = 4$ control at every depth, and $n_{\text{heads}} = 8$ sits 0.026–0.091 nats *below* it at every depth – consistently signed across all five depths and both seeds, not the exact invariance the parameter-counting argument predicts. The effect is real but modest (a third to two-thirds the size of the depth effect itself over this range) and monotone in head count (more heads, lower loss, at fixed width and budget): splitting the same $\dim \times \dim$ projections into more, narrower heads changes something the pure per-block parameter count does not capture – plausibly an optimization or representational effect of head width itself, not a capacity effect, since the quantization model’s own placement of all architecture-dependence in the capacity term predicts no effect here at all. We report this as a genuine, reproducible complication for the simple architecture-independent- c story, not as grounds to discard it: both tests here are quick, single-seed-pair, uncorrected checks, several rungs below the rigor of the retained 16-depth fit, and neither the confounded width test nor the small head-count effect bears on the leading-order identification of c itself (§5), which does not depend on either experiment.

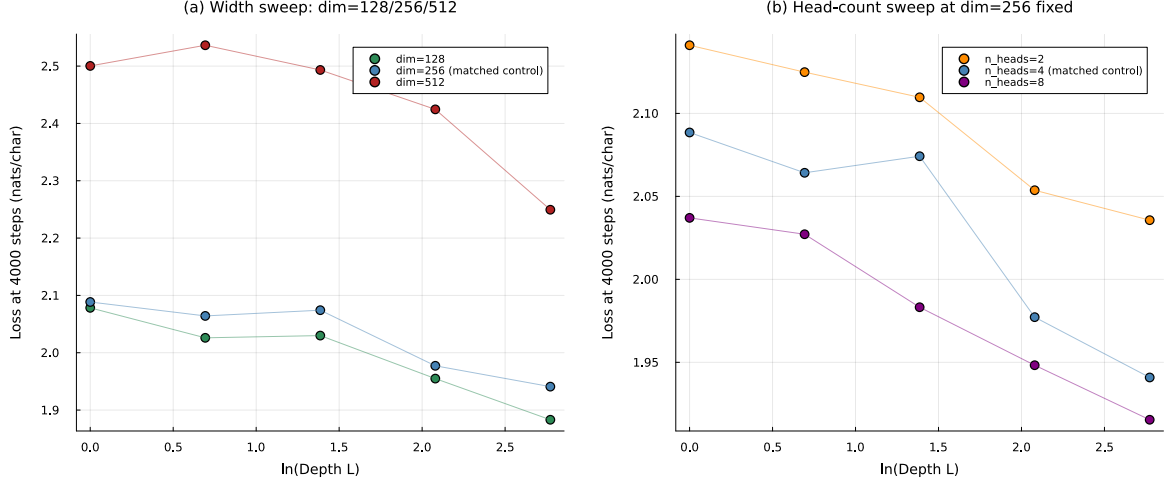


Figure 7: The two discriminating experiments, $L \in \{1, 2, 4, 8, 16\}$, 2 seeds, 4000 steps, identical convention throughout (no truncation-bias correction). (a) Width sweep: dim= 128 tracks the dim= 256 control, but dim= 512 is uniformly elevated – a training-budget confound, not the predicted floor shift. (b) Head-count sweep at dim= 256 fixed: a small, monotone, consistently-signed departure from the predicted exact invariance.

Since a_1, a_2, a_3 depend on only two unknowns (c, μ) , eliminating them gives a **parameter-free, falsifiable constraint**:

$$\frac{a_1 a_3}{a_2^2} = -\frac{1}{3},$$

confirmed by direct computation (an independent, dependency-free implementation of ψ via recursion and the Stirling asymptotic series) to converge to exactly $-1/3$ as $\mu L \rightarrow \infty$ (-0.2916 at $\mu=1$, -0.3322 at $\mu=8$, -0.3333 at $\mu \geq 50$).

Generalizing the Zipf exponent to $1 + \varepsilon$ replaces the harmonic sum with the Hurwitz zeta function, $a(L) = a_\infty + c[\zeta(1+\varepsilon, \mu L+1) - \zeta(1+\varepsilon, K_{\max}+1)]$ (well-defined without a cutoff for $\varepsilon > 0$; $\varepsilon = 0$ is the singular boundary case requiring one). Expanding $[(\mu L)^{-\varepsilon} - 1]/\varepsilon$ in ε at fixed L gives $-\ln L$ at zeroth order: the observed indistinguishability of a pure log fit from a power-law fit with a small exponent (§5: $q \rightarrow 0$) is not a numerical coincidence but the same function evaluated two different ways, coincident whenever $\varepsilon \ln L \ll 1$ – exactly the regime $L \leq 16$, $\varepsilon \approx 0.076$ falls into.

Testing the constraint – a first attempt that self-destructs under scrutiny. Extending the weighted fit of §5 with a_2/L and a_3/L^2 terms yields a nominally favorable $\Delta\text{AIC} = -10.1$, but four robustness checks (residual diagnostics, leave-one-out, and a cluster bootstrap resampling depths rather than only within-depth points) show this is carried entirely by one high-leverage point ($L=2$, whose two pooled data sources already disagree by 1.6σ): removing it alone drops the nested-model significance from $p=0.0009$ to $p=0.26$, and the extended model is selected in only 50% of bootstrap resamples, with a_1 unstable enough to change sign. The nominal $a_1 a_3 / a_2^2 \approx -0.124$ is not interpretable, being built from coefficients just shown not to be robust.

A dedicated, pre-registered campaign – and an inconclusive verdict with a measured cause. We then ran the test this diagnosis calls for. The design decision is where *not* to spend compute: the correction terms carry essentially no curvature signal at large depth (the a_3/L^2 term is $\sim 3 \times 10^{-5}$ nats at $L=16$), so extending to $L = 32, 64$ – the natural-sounding fix – would refine only a_1 , which is already robust. All identifiable signal lives at shallow depth, and so does the known trap: the truncation bias of a 4000-step fit ($+0.09$ nats at $L=2$, §5) is roughly three times the sought $a_2/L + a_3/L^2$ signal and opposite in sign – very plausibly what manufactured the nominal $\Delta\text{AIC} = -10.1$ above. The campaign therefore ran 36 fresh runs: four seeds at each of nine depths, 20,000 steps per run at the signal-bearing depths $L \in \{1, \dots, 6\}$ and 10,000 at the anchor depths $L \in \{8, 12, 16\}$, with full validation curves so each $a(L)$ comes from a per-depth curve fit pooled over seeds, not a final-loss summary. Verdict criteria were fixed before fitting: (1) robust selection of the extended model ($\Delta\text{AIC} < -6$ overall, under every leave-one-out removal, and in

$\geq 90\%$ of cluster-bootstrap resamples); (2) stable coefficient signs ($a_1 < 0$, $a_2 < 0$, $a_3 > 0$ in $\geq 95\%$ of resamples); (3) bootstrap 95% interval of $R = a_1 a_3 / a_2^2$ overlapping the theoretical band $[-0.333, -0.29]$. All three fail: $\Delta\text{AIC} = -5.4$, leave-one-out selection fails, the extended model is selected in 60% of resamples with signs stable in 72%, and $R \in [-0.19, 0.57]$ spans zero and misses the band entirely (Figure 8). The nominal point estimate, $R = -0.125$, is again not interpretable, and we do not report it as a measurement. The verdict has a measured cause, not merely a description: even if the theory were exactly true with $c \approx 0.10$ and $\mu \in [0.7, 2]$, the component of $a_2/L + a_3/L^2$ orthogonal to the $\{1, \ln L\}$ basis over these nine depths – the only part a nested-model comparison can see – peaks at 0.005–0.011 nats, and the expected weighted χ^2 gain is 1.6–6.5, below the ≈ 10 the campaign’s own selection threshold requires. **The theory delivers an exact, falsifiable prediction; this class of experiment, at this scale, cannot decide it even in the theory-favorable case.** The constraint remains open, and the requirement is now quantified: a further two-to-threefold tightening of $a(L)$ at $L \leq 6$ (of order ten times more seed-runs at these budgets, or variance-reduced evaluation) – not deeper models, not more depths. Every algebraic reformulation we checked (exact digamma recurrences that eliminate a_∞ and $K_{\max}$ entirely, ratios of consecutive differences, external anchors on the floor or on total quanta) either collapses back onto the same shallow-depth curvature this section already extracts, or amplifies its noise rather than reducing it: $-1/3$ is tied to this specific extraction at this scale, not to any one way of computing it. The more promising route we did not take here changes the *observable* rather than the algebra – a per-character loss distribution at a fixed checkpoint (no temporal extrapolation, hence none of this section’s dominant error source) can test whether quantization is occurring at all, and a corpus-size sweep at fixed depth measures the Zipf exponent ε directly on the data axis the quantization model actually predicts it for, rather than through $a(L)$ ’s curvature. Both trade $-1/3$ itself for a more measurable invariant of the same underlying model.

What the campaign did measure: a non-monotone anomaly in $a(L)$ at $L = 4$ –5, of training-dynamics origin. The nine pooled floor estimates are precise enough to reveal structure the 16-depth, mostly single-seed fit could not: $a(L)$ is not monotone, with $a(4) = 1.463$ and $a(5) = 1.451$ sitting above $a(3) = 1.442$ before the decline resumes at $a(6) = 1.424$ (within-fit standard errors 0.005–0.006, which understate seed-level variation, so we attach no significance level). Two independent trajectory diagnostics locate the cause in training dynamics rather than capacity: the plateau-escape time (first crossing of 2.2 nats) is latest at $L=4$ (3,250 steps on average, against 2,500 at $L=16$, decreasing monotonically beyond $L=4$), and the seed-to-seed spread of final loss peaks in the same shallow window (up to ± 0.03 nats at $L \in \{2, 4, 5\}$, against ≤ 0.005 at $L \in \{1, 6, 8\}$). The quantization model makes claims about the floor, not about how fast optimization reaches it; a depth-localized escape-time effect is outside the model – and is exactly the kind of dynamics-floor entanglement that any finite-budget estimate of $a(L)$ at these depths inherits.

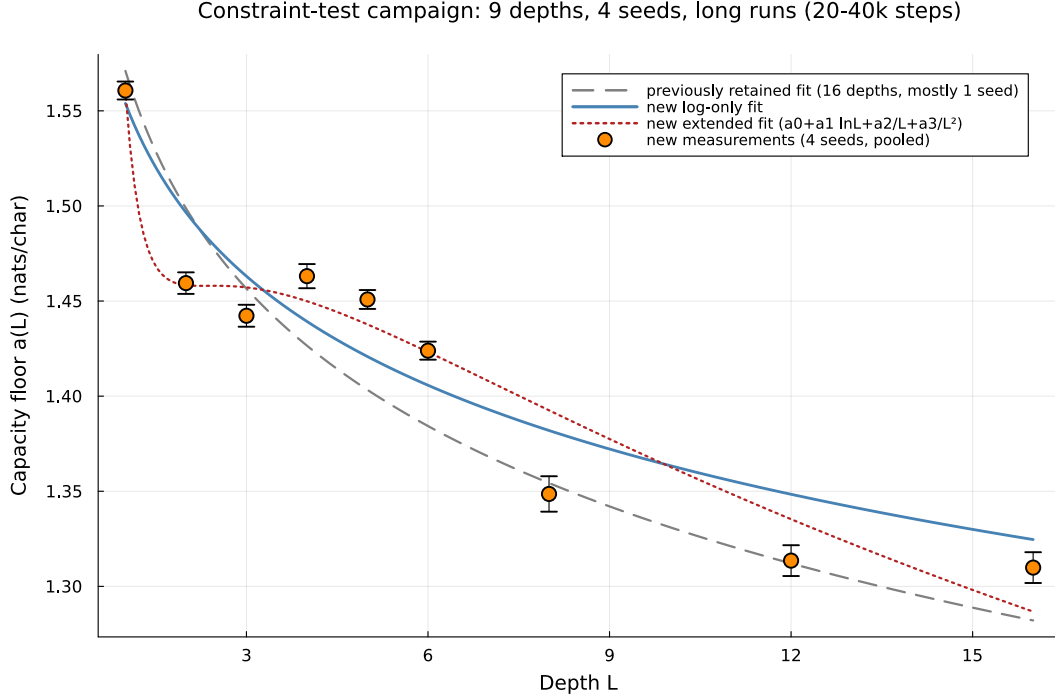


Figure 8: The constraint-test campaign: nine pooled $a(L)$ estimates (four seeds each, long runs), against the previously retained 16-depth fit (dashed), the campaign’s own log-only refit (solid), and the extended fit with $a_2/L + a_3/L^2$ terms (dotted). The extended fit fails all three pre-registered selection criteria, and the visible non-monotone excursion at $L = 4-5$ is of training-dynamics origin, not a floor effect.

7 Validation Against Eighteen Measured Schedules

Chaining the canonical curves through the effective-step rule, with graft boundaries read directly from measured schedules (no resimulation, no parameter fit to this data), predicts all 18 schedules from §2 (Table 1) with $R^2=0.764$, $\text{RMSE}=0.0849$ nats – correctly reproducing both regime reversals (growth wins at equal FLOPs, fixed wins at equal steps) and each campaign’s coarse ordering. Validated instead against every individual point on every trajectory (1536 points, not just final losses), $R^2=0.956$, $\text{RMSE}=0.059$. Residuals are structured, not noise: grafted arms underperform the prediction by 0.04–0.17 nats, correlated with the number of grafts ($r=0.43$, slope $\approx +0.005$ nat/graft) – a real, unmodeled transient cost per graft that the effective-step rule, exact only at the instant of grafting, does not capture. This is the main falsifiable hypothesis this validation surfaces beyond the constraint of §6. This RMSE should be read against the systematic floor of the model’s own ingredients: the canonical $a(L)$ estimates shift by 0.02–0.09 nats under fit-window choices alone (§6), so an aggregate prediction error of ≈ 0.08 nats is roughly as small as these inputs permit.

Campaign	Schedule	Measured	Predicted	Error
Equal FLOPs	fixed-4 (20k steps)	1.6535 ± 0.0267	1.6159	-0.038
Equal FLOPs	2→3→4	1.6032 ± 0.0107	1.5852	-0.018
Equal FLOPs	1→2→4	1.6366 ± 0.0025	1.5746	-0.062
Equal FLOPs (equal-share)	fixed-16 (3k steps)	2.0562 ± 0.0059	2.1167	+0.061
Equal FLOPs (equal-share)	1→16	1.7191 ± 0.0035	1.6599	-0.059
Equal FLOPs (equal-share)	4 stages	1.7838 ± 0.0082	1.6849	-0.099
Equal FLOPs (equal-share)	8 stages	1.8151 ± 0.0352	1.7125	-0.103
Equal FLOPs (equal-share)	16 stages	1.8053 ± 0.0543	1.7366	-0.069
Equal FLOPs (1/ L -weighted)	fixed-16 (3k steps)	2.0723 ± 0.0049	2.1167	+0.044
Equal FLOPs (1/ L -weighted)	1→16	1.6542 ± 0.0139	1.6432	-0.011
Equal FLOPs (1/ L -weighted)	4 stages	1.7520 ± 0.0003	1.6384	-0.114
Equal FLOPs (1/ L -weighted)	8 stages	1.8082 ± 0.0223	1.6356	-0.173
Equal FLOPs (1/ L -weighted)	16 stages	1.7593 ± 0.0213	1.6423	-0.117
Equal steps (4k)	fixed-16	1.9570 ± 0.0086	1.9975	+0.040
Equal steps (4k)	1→16	2.0619 ± 0.0541	2.0115	-0.050
Equal steps (4k)	4 stages	2.1195 ± 0.0278	1.9996	-0.120
Equal steps (4k)	8 stages	2.1169 ± 0.0444	1.9971	-0.120
Equal steps (4k)	16 stages	2.0233 ± 0.0311	1.9962	-0.027

Table 1: All 18 measured schedules (mean of seeds) vs. predictions chained from the fitted canonical curves through the effective-step rule – $R^2=0.764$, RMSE= 0.0849 nats.

Limitations. The anomaly that 16 stages beats 8 at equal FLOPs is reproduced in sign only at equal steps, not at equal FLOPs, and remains unexplained – reported, not forced. $a(L)$ ’s log form cannot extrapolate far past $L=16$ (it would turn negative near $L \approx 3.5 \times 10^6$, with no saturation yet visible on the tested range), and the envelope argument of Theorem 2 is verified only on that same range: under the fitted log form, a is convex in $\kappa(L) = \sqrt{Lb(L)}$, so the theoretically optimal equal-FLOPs schedule is graded, concentrated on depths 1–3 (1.626 predicted vs. 1.643 for the best single jump) – a 0.017-nat gap now known to sit below the measured systematic floor of the fitted inputs themselves ($a(L)$ moves by 0.02–0.09 nats across defensible fit windows, §6; schedule-level RMSE is 0.085). The envelope argument of the theorem stands; its numeric prediction at this scale is not testable against these fits, only against a dedicated multi-seed run of the graded schedule. All of this paper holds at one scale (a ~ 5 –20M-parameter char-LM, batch size 1, one task); generalizing across scale is exactly what a larger, GPU-scale replay of the winning schedule would test.

8 Discussion and Conclusion

Growth beats fixed-depth training at equal compute but loses at equal step count, and among schedules actually tested, the coarsest wins at equal compute; we do not leave this as a measured curiosity, nor as a curiosity of an undertuned baseline – a cosine learning-rate decay on the fixed-depth-16 baseline closes roughly half the gap but leaves the ranking unchanged (§2), a pre-registered check run specifically because it is the first objection this result invites. Three theorems, proven by comparison arguments on an ODE model of the canonical loss curve, pin the mechanism to a single quantity: a cost-vs-capacity threshold in which cost enters sub-linearly and the capacity gap linearly, exactly why the same pair of schedules can flip which one wins depending on which resource is held fixed. The domination theorem’s necessity direction depends on a single-crossing lemma to split correctly into the two cases a non-decreasing growth schedule actually allows – training at a shallower depth and grafting up, or matching the fixed baseline until a crossing loss and only then grafting deeper – rather than one construction serving both.

We push further still, deriving the fitted capacity floor’s logarithmic form not as the best curve tried but as the exact solution of a quantization-model recurrence, closed in the digamma function, whose asymptotic expansion yields a parameter-free constraint on its own coefficients ($a_1 a_3 / a_2^2 = -1/3$). We tested it with a dedicated, pre-registered 36-run campaign; the verdict is inconclusive on all three pre-registered criteria, and the campaign quantifies why: even in the theory-favorable case, the detectable signal at this scale sits below the model-selection threshold. The constraint stays open with its detectability limit now measured – reported

rather than either forced to a verdict or quietly dropped. The quantization model’s own quanta-per-layer rate μ fares similarly: it is not identified by curve-fitting on current data at all, for a precisely diagnosed reason ($\ln \mu$ is absorbed into the model’s intercept to within 98.6% correlation). We tested the cheap fix this diagnosis suggested – ten extra seeds at $L = 2-3$ – and it resolves a different question than the one asked: it independently confirms the excluded $L=3$ outlier was a genuine single-seed artifact (0.91σ from trend on fresh data, against 6.68σ before), but leaves μ itself unidentified, revealing that the binding constraint was run *length*, not seed count, at these depths (§6) – reporting the correction to our own proposed fix rather than silently dropping it. Pooling every from-scratch measurement in this paper – 3,028 points across all sixteen depths, more than double the original fit – moves the point estimate but not the verdict: a depth-cluster bootstrap still spans $[0.012, 100]$, confirming that the limit is which depths are informative, not how much data is collected. The complete model, fit to sixteen depths and validated against eighteen independently measured schedules with no parameter re-fit, explains both regime reversals at once; its residuals are structured rather than noise, pointing to an unmodeled transient cost per graft as the next concrete question this validation raises.

NEURODSL is open-source at <https://github.com/nevermind78/NeuroDSL>.

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